

## Program 4

### Integrate Git with Docker for Source Controlled Application Builds

#### Description

The problem statement involves integrating Git with Docker to enable source controlled application builds. This ensures automated, version controlled and reproducible Docker image directly from a Git repository.

#### Project Structure

Program4  
→ Dockerfile  
→ [index.js](#)  
→ package.json  
→ package-lock.json

Step 1: Make a folder program4 and move into the folder

```
mkdir program4  
cd program4
```

Step 2: Create a file → package.json and execute the following command

```
npm init
```

```
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program4/Devops_program4$ cat package.json  
{  
  "name": "program4",  
  "version": "1.0.0",  
  "description": "",  
  "main": "index.js",  
  "scripts": {  
    "test": "echo \"Error: no test specified\" && exit 1"  
  },  
  "keywords": [],  
  "author": "",  
  "license": "ISC",  
  "type": "commonjs",  
  "dependencies": {  
    "express": "^5.1.0"  
  }  
}
```

Step 3: Create a file [index.js](#)

```
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program4/Devops_program4$ cat index.js
const express = require('express');
const app = express();
const port = 3000;

app.get('/', (req,res) =>{
    res.send("Hello from Git + Docker integration");
});

app.listen(port, () => {
    console.log('server running at http://localhost:${port}');
});
```

#### Step 4: Create the Dockerfile

```
jeevanc@jeevanc-IdeaPad-Slim-3-15AMN8:~/Devopslab/program4/Devops_program4$ cat Dockerfile
FROM node:18 AS build
WORKDIR /app
COPY package*.json ./
RUN npm install
COPY . .

FROM node:18-slim
WORKDIR /app
COPY --from=build /app /app
EXPOSE 3000
CMD ["node", "index.js"]
```

#### Step 5: Create a New Repo on Github

#### Step 6: Get a personal access token (PAT)

Go to profile -> Developer settings -> Personal Access Token -> Token(classic) and generate a token

#### Step 7: Initialize Git Repository locally

```
git init
git add .
git commit -m "Initial commit"
git branch -M main
```

#### Step 8: Link to Remote Repository

```
git remote add origin https://github.com/JeevanC37/program.git
```

#### Step 9: Push code to Github

```
git push -u origin main
```

```

jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/DevopsLab/program4/Devops_program4$ git init
Reinitialized existing Git repository in /home/jeevanc/DevopsLab/program4/Devops_program4/.git/
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/DevopsLab/program4/Devops_program4$ git add .
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/DevopsLab/program4/Devops_program4$ git commit -m "initial commit"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/DevopsLab/program4/Devops_program4$ git branch -M main
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/DevopsLab/program4/Devops_program4$ git remote add origin https://github.com/JeevanC37/Devops_program4.git
error: remote origin already exists.
jeevanc@jeevanc-IdeaPad-Slin-3-15AMN8:~/DevopsLab/program4/Devops_program4$ git push -u origin main
Username for 'https://github.com': JeevanC37
Password for 'https://JeevanC37@github.com':

```

The screenshot shows a web browser displaying the GitHub repository page for 'Devops\_program4' by user 'JeevanC37'. The repository is public and has 1 branch (main) and 0 tags. The commit history shows a single commit 'first commit' by 'JeevanC37' 4 hours ago, with 2 commits in total. The file list includes 'node\_modules', 'Dockerfile', 'README.md', 'index.js', 'package-lock.json', and 'package.json', all marked as 'first commit' 4 hours ago. The README section is visible, titled 'Devops\_program4'. On the right sidebar, there are sections for 'About' (no description), 'Releases' (no releases published), 'Packages' (no packages published), 'Languages' (JavaScript 58.5%, Dockerfile 41.5%), and 'Suggested workflows' (SLSA Generic generator).